

BANJA LUKA INTL, Bosnia and Herzegovina

WMO: 132420

Lat: 44.941N

Lon: 17.298E

Elev: 122

StdP: 99.87

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.0	-8.0	-13.8	1.2	-10.0	-10.2	1.6	-6.9	7.1	5.9	6.2	6.1	1.2	160	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	34.8	21.8	32.9	21.7	31.0	21.0	23.5	30.8	22.6	30.1	21.7	28.9	2.6	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.0	26.7	20.2	15.1	25.7	19.2	14.2	24.6	70.3	30.7	67.1	30.1	64.2	29.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.5	DB	-15.1	37.1	4.5	2.3	-18.4	38.7	-21.0	40.1	-23.5	41.4	-26.8	43.1	(4)
(5)			WB	-15.5	25.3	4.4	0.9	-18.6	25.9	-21.2	26.4	-23.7	26.9	-26.8	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	1.2	2.7	7.5	12.8	16.5	20.7	22.7	22.1	17.4	12.3	7.3	2.3	(6)	
	DBStd	8.57	4.97	5.12	4.77	3.47	3.62	3.63	3.09	3.20	3.66	3.93	4.33	4.56	(7)	
	HDD10.0	963	273	206	102	14	2	0	0	0	0	22	100	243	(8)	
	HDD18.3	2658	531	437	336	169	77	18	3	7	61	191	330	498	(9)	
	CDD10.0	1755	1	2	24	97	204	321	393	376	221	92	20	3	(10)	
	CDD18.3	410	0	0	0	2	22	90	139	123	32	3	0	0	(11)	
	CDH23.3	4623	0	0	2	59	292	967	1545	1342	369	46	1	0	(12)	
	CDH26.7	1929	0	0	0	9	70	379	716	625	125	4	0	0	(13)	
(14) Wind	WSAvg	1.8	1.6	1.8	2.2	2.0	2.0	1.9	1.8	1.7	1.7	1.6	1.6	1.6	(14)	
(15) Precipitation	PrecAvg	1064	71	66	81	95	106	106	87	90	106	90	94	88	(15)	
	PrecMax	1617	136	116	150	215	206	205	219	317	264	176	204	177	(16)	
	PrecMin	630	12	15	11	18	45	55	36	9	31	11	9	4	(17)	
	PrecStd	217	34	29	33	48	51	40	44	77	65	48	46	48	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	18.8	23.1	28.1	31.0	34.2	37.0	37.2	33.2	27.0	22.2	18.1	(19)	
		MCWB	10.7	11.0	12.6	17.0	20.2	22.8	21.9	21.2	20.4	18.0	14.9	12.4	(20)	
	2%	DB	13.2	15.2	20.9	24.9	28.9	32.2	34.9	34.8	30.1	24.8	19.2	15.1	(21)	
		MCWB	9.0	9.4	12.0	15.3	19.7	22.1	22.0	21.3	19.4	16.9	13.4	10.6	(22)	
	5%	DB	10.1	12.9	18.2	22.9	26.8	30.9	32.9	32.2	27.2	22.0	16.9	12.0	(23)	
		MCWB	7.7	8.3	11.0	14.3	18.0	21.4	22.0	21.3	18.9	16.1	12.2	8.6	(24)	
	10%	DB	7.8	10.0	16.0	20.8	24.8	29.0	30.9	30.2	25.0	19.1	14.0	8.8	(25)	
		MCWB	6.2	7.3	9.9	13.3	17.4	20.5	21.0	20.8	17.9	14.7	11.1	7.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	11.4	14.0	17.6	21.2	24.4	25.0	24.1	21.8	18.9	15.6	12.9	(27)	
		MCDB	15.3	17.1	20.6	26.1	28.8	31.5	32.1	31.1	28.3	25.2	20.8	17.6	(28)	
	2%	WB	9.4	10.0	12.6	16.0	20.1	23.2	23.5	23.0	20.2	17.5	14.0	10.7	(29)	
		MCDB	12.7	14.4	18.9	22.8	27.1	30.4	30.9	30.5	27.7	22.9	18.4	14.6	(30)	
	5%	WB	7.8	8.6	11.5	15.0	19.0	22.2	22.6	22.2	19.4	16.4	12.8	9.0	(31)	
		MCDB	9.9	12.3	17.3	20.7	25.1	28.7	30.5	29.8	26.1	21.1	16.2	11.9	(32)	
	10%	WB	6.0	7.2	10.3	14.0	17.8	21.1	21.7	21.4	18.5	15.3	11.3	7.1	(33)	
		MCDB	7.1	9.3	15.3	19.3	23.2	27.4	29.0	28.4	23.8	19.0	14.1	8.3	(34)	
(35) Mean Daily Temperature Range	MDBR		7.2	8.2	10.7	12.2	11.6	11.7	12.7	12.9	11.6	10.6	8.7	6.8	(35)	
	5% DB	MCDBR	11.6	14.0	16.0	17.1	15.5	15.1	16.6	17.5	16.0	15.3	14.0	12.4	(36)	
		MCWBR	8.1	8.7	8.6	8.4	7.2	6.4	6.0	6.3	7.0	8.6	8.7	8.5	(37)	
	5% WB	MCDBR	11.1	12.7	13.8	14.1	13.9	13.7	14.3	14.6	13.9	13.8	12.8	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.332	0.361	0.395	0.448	0.436	0.435	0.435	0.436	0.421	0.390	0.353	0.332	(40)	
	taud		2.388	2.303	2.247	2.118	2.202	2.247	2.243	2.249	2.264	2.346	2.402	2.402	(41)	
	Ebn at Noon		771	810	830	813	837	838	832	816	794	767	737	729	(42)	
	Edh at Noon		76	99	120	148	140	135	134	128	117	94	75	69	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.27	1.95	3.14	4.28	5.31	6.00	6.11	5.44	3.72	2.53	1.47	1.05	(44)	
	RadStd		0.19	0.29	0.43	0.52	0.47	0.50	0.43	0.60	0.46	0.30	0.18	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.50	N/A	N/A	N/A	+0.35	N/A	N/A	-120	N/A	N/A

Nomenclature: See separate page